KLE Technological University | Creating Value, Leveraging Knowledge

DEPARTMENT OF COMPUTER APPLICATIONS

MINOR PROJECT

Restaurant Searching And Ranking System

Submitted by:

Akash Hadapad- 01FE24ICA063

Ramling Kalal – 01FE24ICA059

INTEGRATED MCA

DIV: F

Submitted to:

2026-2027

1.Introduction

The Restaurant Searching and Ranking System is a web-based application that helps users find suitable restaurants according to their needs and preferences. Nowadays, people often face difficulties when choosing a restaurant because there are many restaurants available in different locations, with different food varieties, prices, and ratings. Searching for a good restaurant manually can take a lot of time.

The main purpose of this project is to make restaurant searching easier, faster, and more convenient. The system allows users to search for restaurants based on location, food type, price range, ratings, and customer reviews. It displays a list of matching restaurants and ranks them according to the selected preferences.
The system provides useful information about each restaurant, such as its name, location, cuisine, price range, ratings, and reviews. This information helps users compare different restaurants and choose the one that best suits their requirements.

2.Objectives

The main objectives of the Restaurant Searching and Ranking System are:

1. To help users search for restaurants easily.
2. To allow users to find restaurants based on location and food type.
3. To provide information about restaurant prices, ratings, and reviews.
4. To rank restaurants according to ratings and user preferences.
5. To help users compare different restaurants before making a decision.
6. To reduce the time required to find a suitable restaurant.
7. To provide a simple and user-friendly interface.
8. To allow administrators to add, update, and delete restaurant information.
9. To improve the restaurant selection process by displaying relevant search results.

3.Methodology

- The methodology explains the steps used to develop the Restaurant Searching and Ranking System.

- **5.1 Data Collection**
- Collect restaurant details such as name, location, food type, price, ratings, and reviews.

- **5.2 Database Design**
- Store restaurant information in a database for easy access and management.

- **5.3 User Search**
- Allow users to search for restaurants by name, location, or food type.

- **5.4 Filtering**
- Provide filters based on price range, food type, and ratings.

- **5.5 Ranking**
- Rank restaurants according to their ratings and how well they match the user's preferences.

- **5.6 Display Results**
- Display matching restaurants with their details, ratings, and reviews.

- **5.7 Testing**
- Test the system to ensure that searching, filtering, ranking, and displaying results work correctly.

4.Modules

- The Restaurant Searching and Ranking System consists of the following modules.
- **1. User Module**
- This module allows users to access the system and search for restaurants. It provides a simple interface for viewing restaurant information and comparing available options.

- **2. Restaurant Search Module**
- This module allows users to search for restaurants using restaurant names, locations, food types, or keywords. It retrieves matching restaurant information from the database.

- **3. Filter Module**
- This module helps users narrow down search results according to their preferences. Users can filter restaurants by cuisine, price range, ratings, and other available details.

- **4. Restaurant Ranking Module**
- This is one of the main modules of the system. It ranks restaurants according to selected factors, such as customer ratings, price suitability, location match, and search relevance. This helps users identify suitable restaurants more easily.

- **5. Restaurant Details Module**
- This module displays detailed information about a selected restaurant. The information may include the restaurant name, address, food type, estimated cost, ratings, reviews, and contact details.

5.Step-by-Step Workflow

1. The working process of the Restaurant Searching and Ranking System is explained below.

2. **Step 1: Open the Application**

The user opens the Restaurant Searching and Ranking System through a web browser.

3. **Step 2: Enter Search Details**

The user enters a restaurant name, location, cuisine, or other search keywords into the search box.

4. **Step 3: Retrieve Restaurant Information**

The system searches the database and retrieves restaurants that match the user's search requirements.

5. Step 4: Apply Filters

The user can apply filters such as price range, cuisine type, and minimum rating to narrow down the results.

6.Tools and Technologies

The following tools and technologies can be used to develop the Restaurant Searching and Ranking System.

- **1. HTML**
- HTML is used to create the structure of web pages.
- **2. CSS**
- CSS is used to design web pages and make them attractive.
- **3. JavaScript**
- JavaScript is used to make the website interactive.
- **4. Node.js and Express.js**
- These are used to handle user requests and process restaurant search information.
- **5. MongoDB**
- MongoDB is used to store restaurant details, ratings, and reviews.
- **6. Visual Studio Code**
- Visual Studio Code is used to write and manage the project code.
- **7. Web Browser**
- Google Chrome or Microsoft Edge is used to run and test the website.

7. Features and Specifications

- The system provides useful features to help users find suitable restaurants easily.
- **1.1 Restaurant Search**
- Users can search for restaurants by name, location, or food type.
- **1.2 Restaurant Filtering**
- Users can filter restaurants by food type, price range, and ratings.
- **1.3 Restaurant Ranking**
- The system ranks restaurants based on ratings and user preferences.
- **1.4 Restaurant Details**
- The system displays restaurant names, locations, food types, and prices.
- **1.5 Ratings and Reviews**
- Users can view customer ratings and reviews before choosing a restaurant.
- **1.6 Admin Management**
- The administrator can add, update, and delete restaurant details.
- **1.7 User-Friendly Interface**
- The system provides a simple interface for easy searching and viewing results.
- **1.8 Database Management**
- The database stores restaurant details, ratings, and reviews.

8. Hardware Requirements

The minimum hardware requirements are:

1. **Processor:** Intel Core i3 or equivalent.
2. **RAM:** Minimum 4 GB.
3. **Storage:** 10 GB free space.
4. **System:** Desktop or laptop.
5. **Internet:** Required for online access.

9. Software Requirements

- **3.1 Operating System**
- Windows 10/11 or Linux.

- **3.2 Frontend**
- HTML, CSS, and JavaScript for creating web pages.

- **3.3 Backend**
- Node.js and Express.js for handling application logic.

- **3.4 Database**
- MongoDB or MySQL for storing restaurant information.

- **3.5 Code Editor**
- Visual Studio Code for writing code.

- **3.6 Web Browser**
- Google Chrome, Microsoft Edge, or Mozilla Firefox.

10. References

- **HTML Documentation – MDN Web Docs**
- **CSS Documentation – MDN Web Docs**
- **JavaScript Documentation – MDN Web Docs**
- **Node.js Documentation – Node.js**
- **Express.js Documentation – Express.js**
- **MongoDB Documentation – MongoDB**
- **MySQL Documentation – MySQL**
- **React.js Documentation – React**

11. Conclusion

- The Restaurant Searching and Ranking System helps users find restaurants based on location, food type, price, and ratings. It ranks restaurants according to user preferences and displays useful details and reviews. The administrator can manage restaurant information. Overall, the system makes restaurant searching easier and saves users time.